preface

Machine learning, AI, large language models (LLMs), and generative AI are technologies that have fundamentally reshaped our digital landscape, opening up new avenues for innovation. Their primary appeal lies in their ability to facilitate multimodal interactions—combining text, images, and videos—between humans and machines in a remarkably natural way. It's as simple and intuitive as asking a friend or colleague for help. These applications are no longer confined to science fiction; they're gaining new practical uses every day as technological advancements make them increasingly accurate and lifelike.

As a seasoned software engineer deeply involved in building AI-powered products and supporting headless WordPress solutions on Next.js and React, I've had a front-row seat to this transformation. The imminent access to a plethora of open source tools and initiatives, coupled with the sheer accessibility of LLMs and related AI services, sparked an idea: to bridge the gap for web developers eager to harness this technology without needing a deep dive into machine learning theory or Python.

This book was born from a clear vision: AI and its applications are powerful new tools that every developer must be able to use to deliver exceptional value to their users and customers. It's designed primarily as a practical guide, but I've also placed a strong emphasis on explaining the core concepts behind building full-stack generative web applications.

I've specifically chosen JavaScript, Next.js, Vercel, and LangChain.js because they form a powerful, accessible, and incredibly relevant stack for creating generative AI experiences today. My hope is that this book will become your essential companion, making your journey into building AI-driven web applications both enjoyable and profoundly educational.